Algebraic Structures

Algebraic Structures	Definition
Group $(G, \star)$	• $\star$ Closure: $\forall a, b \in G, a \star b \in G$ • Associativity: $\forall a, b, c, (a \star b) \star c = a \star (b \star c)$ • Identity: $\forall a, a \star e = a = e \star a$ • Inverses: $\forall a, \exists b, a \star b = e = b \star a$ • (Commutativity for commutative group): $\forall a, b, a \star b = b \star a$
Ring $(R, +, \cdot)$	• $(R, +)$ is a commutative group • $\cdot$ Closure, associativity, identity (commutativity for commutative ring) • Left/right distributivity: $\forall a, b, c, a \cdot (b + c) = (a \cdot b) + (a \cdot c), (a + b) \cdot c = (a \cdot c) + (b \cdot c)$
Integral Domain	commutative right with $1 \neq 0$ and no zero divisors $ab = 0 \Rightarrow a = 0$ or $b = 0$
Field F	commutative right with $1 \neq 0$ and every $x \neq 0$ has a mult inverse
F-vector space $(V, +, s)$	• F is a field • $(V, +)$ is commutative group • Scalar multiplication $s : F \times V \to V$: $\forall c, d \in F, v, w \in V$ closed: $s(c, v) = cv \in V$ distributivity of addition: $(c + d)v = cv + dv, c(v + w) = cv + cw$ associativity of scalar multiplication: $c(dv) = (c \cdot d)v$ identity: $1_F v = v$
F-algebra $(A, +, \cdot, s)$	• $(A, +, \cdot)$ is a ring • $(A, +, s)$ is an F-vector space • For $c \in F, v, w \in A, c(v \cdot w) = (cv) \cdot w = v \cdot (cw)$

Sub-Algebraic Structures	Definition
Subgroup H of $(G, \star)$	closure, identity, inverses **One-Step Subgroup Test**: H is a subgroup $\Leftrightarrow \forall a, b \in H, ab^{-1} \in H$
Cyclic Subgroup	$\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$ is always a subgroup of G
Normal Subgroup	closure under conjugation: $\forall g \in G, h \in H, ghg^{-1} \in H$
Subring S of $(R, +, \cdot)$	$0_R, 1_R \in S$ and $\forall a, b \in S, a + b, -a, a \cdot b \in S$
Ideal I of $(R, +, \cdot)$	S is an additive subgroup s.t. $\forall a \in R, s \in S, a \cdot s, s \cdot a \in S$ (not necessarily a subring) $1_R \in I \Leftrightarrow I$ is a subring
Subfield K of F	subring s.t. $\forall a \neq 0 \in K, a^{-1} \in K$
Subspace W of V	• $0_V \in W$ • $\forall v, w \in W, v + w \in W$ • $\forall c \in F, w \in W, cw \in W$
Subalgebra B of A	both a subring of the ring A and a subspace of the vector space A

Homomorphism	Definition
Goup Homomorphism	• $T : G \to K$ s.t. $T(x \star y) = T(x) \star T(y)$ • $T(e_G) = e_K, T(x^{-1}) = T(x)^{-1}, T(x^n) = T(x)^n$
Ring Homomorphism	• $T : R \to S$ s.t. $T(x + y) = T(x) + T(y)$ • $T(x \cdot y) = T(x) \cdot T(y)$ • $T(1_R) = 1_S$
F-linear map (vector space homomorphism)	• $T : V \to W$ s.t. $T(u + v) = T(u) + T(v)$ • $T(cv) = cT(v)$
Algebra Homomorphism	both a ring and vector space homomorphism

If T is a group homomorphism, the kernel of T is $\ker(T) = \{x \in X : T(x) = 0_Y\}$ is a normal subgroup of X and $\text{img}(T)$ is a subgroup of Y

T is injective $\Leftrightarrow \ker(T) = \{0_X\}$; T surjective $\Leftrightarrow \text{img}(T) = Y$

Theorem (The First Isomorphism Thoerem): $T : A \to B$ is a [group, ring,vector space] homomorphism $\Rightarrow A/\ker(T) \cong \text{img}(T)$ by the isomorphism $T' : A/\ker(T) \to \text{img}(T)$ via $[a \star \ker(T) \mapsto T(a), a + \ker(T) \mapsto T(a), a + \ker(T) \mapsto T(a)]$

Basis

Spanning sets and linearly independent sets:

- S spans $V, S \subseteq T \Rightarrow T$ spans V
- L is linearly independent, $T \subseteq L \Rightarrow T$ is linearly independent $\Rightarrow \varnothing$ is linearly independent
- L is linearly independent $\Rightarrow \exists$ a basis B s.t. $L \subseteq B$
- S spans $V \Rightarrow \exists$ a basis B s.t. $B \subseteq S$
- S spans V, L is linearly independent $\Rightarrow |L| \leq |S|$

Theorem (Rank-Nullity Theorem): Given finite dimensional F-vector space V, W with linear map $T : V \to W$ with null space $N = \ker(T)$ and range $R = \text{img}(T)$, we have $\dim(N) + \dim(R) = \dim(V)$

Matrix Basics

Notation	Definition
e_{ij}	matrix with $e_{ij}(i,j) = 1_F$ and $e_{ij}(k,l) = 0_F$ for $(k,l) \neq (i,j)$
$A_{[i]}$	i-th row of A
$A^{[j]}$	j-th col of A
L_A	vector space homomorphism $F^n \to F^m$ via $v \mapsto Av$

Inverse:

- A invertible $\Leftrightarrow A^T$ invertible: $(A^T)^{-1} = (A^{-1})^T$
- A, B invertible $\Rightarrow AB$ invertible: $(AB)^{-1} = B^{-1}A^{-1}$
- $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Matrix Multiplication:

- $AB(i,j) = \sum_{k=1}^{n} A(i,k)B(k,j)$
- $(AB)^{[j]} = A(B^{[j]})$
- $(AB)_{[i]} = (A_{[i]})B$

Column Space: $\text{Col}(A) = \{Av \mid v \in F^n\} \subseteq F^m$; $\text{colrk}(A) = \dim(\text{Col}(A))$

THEOREM 2.10.2: If V is a k-dim vector space over F then $V \cong F^k$

THEOREM 2.12.1: For $A \in M_{m,n}(F), \operatorname{colrk}(A) \leq \min\{m, n\}$

THEOREM 2.12.2: For $A \in M_{m,n}(F), B \in M_{n,p}(F), \operatorname{colrk}(AB) \leq \min\{\operatorname{colrk}(A), \operatorname{colrk}(B)\}$

THEOREM 2.12.3: If $A \in M_{m,n}(F)$ and $P \in M_m(F)$ invertible then $\operatorname{colrk}(PA) = \operatorname{colrk}(A)$; if $Q \in M_n(F)$ invertible then $\operatorname{colrk}(AQ) = \operatorname{colrk}(A)$

THEOREM 2.12.4: If $A \in \mathrm{GL}_n(F)$ then $\operatorname{colrk}(A) = n$ (full rank)

THEOREM 2.12.5: For $A \in M_{m,n}(F), \operatorname{rowrk}(A) = \operatorname{colrk}(A) = \operatorname{rank}(A)$

THEOREM 2.12.6: If $A \in \mathrm{GL}_n(F)$ then $A_{\text{ech}} = I_n$

THEOREM (INVERTIBLE MATRIX THEOREM): For $A \in M_n(F)$, TFAE:

(1)	A is invertible.
(2a)	$BA = I_n$ for some $B \in M_n(F)$.
(2b)	$AC = I_n$ for some $C \in M_n(F)$.
(3)	$\det(A) \neq 0_F$.
(4)	A is a product of finitely many elementary matrices.
(5)	The reduced row-echelon form A_{ech} is I_n.
(6a)	$\operatorname{rowrk}(A) = n$.
(6b)	The n rows of A form a linearly independent list in F^n.
(7a)	$\operatorname{colrk}(A) = n$.
(7b)	The n columns of A form a linearly independent list in F^n.
(7c)	The range of A has dimension n.
(8a)	The map L_A is surjective (onto).
(8b)	For all $b \in F^n$, there exists $x \in F^n$ with $Ax = b$.
(9a)	The map L_A is injective (one-to-one).
(9b)	For all $x \in F^n$, if $Ax = 0$ then $x = 0$.
(9c)	The null space of A is $\{0\}$.
(10a)	The map L_A is bijective (and hence an isomorphism).
(10b)	For all $b \in F^n$, there exists a unique $x \in F^n$ with $Ax = b$.
(11)	A^T is invertible.
(12)	[when $F = \mathbb{C}$] A^* is invertible.

DETERMINANT

INVERSION: a pair $\{i, j\}$ s.t. $i < j, f(i) > f(j)$ where $f : [n] \to [n]$; $\operatorname{inv}(f) = \#$ inversions and $\operatorname{sgn}(f) = (-1)^{\operatorname{inv}(f)}$

MATRIX DETERMINANT FORMULA:

$$\det(A) = \sum_{f \in S_n} \operatorname{sgn}(f) \prod_{i=1}^{n} A(f(i), i) = \sum_{f \in S_n} \operatorname{sgn}(f) \prod_{i=1}^{n} A(i, f(i))$$

MULTILINEARITY: A function $f : V_1 \times \cdots \times V_n \to W$ for V_i, W vector spaces over F is multilinear if the maps obtained by $v_i \in V_i, \forall i$ but one are all linear

(i, j)-MINOR: $m_{i,j} = \det(A[i \mid j])$ where $A[i \mid j]$ is A without row i and col j

ADJUNCT / CLASSICAL ADJOINT: $A'(i, j) = (-1)^{i+j} m_{i,j}$

Permanent:

$$\operatorname{per}(A) = \sum_{f \in S_n} \prod_{j \in [n]} A(f(j), j) = \sum_{f \in S_n} \prod_{i \in [n]} A(f(i), i) = \operatorname{per}(A^T)$$

Theorem 2.17.1: For any n, the map $\operatorname{sgn} : S_n \to \{-1, 1\}$ is a homomorphism into $\{-1, 1\}$ as a group with multiplication: $\forall f, g \in S_n, \operatorname{sgn}(f \circ g) = \operatorname{sgn}(f)\operatorname{sgn}(g)$

Theorem 2.17.2: The determinant map on rows $\det : (R^n)^n \to R$ is R-multilinear

Theorem (Alternating Property): $A \in M_n(R)$ has two identical rows $\Rightarrow \det(A) = 0$

Theorem (Product Formula): $A, B \in M_n(R), \det(AB) = \det(A)\det(B)$

Theorem (Symmetric Property): If the rows/cols are permuted, the permanent is unchanged <u>(no alternating property or product formula)</u>

Theorem (Laplace Expansion):

$$\det(A) = \sum_{j=1}^{n} A(i, j)(-1)^{i+j} m_{i,j}$$

Theorem (Cauchy-Binet Formula): $A \in M_{m,n}(R), I \subseteq [m], J \subseteq [n], A_{I,J}$ is the $|I| \times |J|$ matrix with restricted map $A : I \times J \to F$

$$\det(AB) = \sum_{k \in \binom{[n]}{m}} \det(A_{[m],k}) \det(B_{k,[m]})$$

Theorem 2.26.1: A, B are square matrices $\Rightarrow \chi_{AB}(x) = \chi_{BA}(x)$

Theorem 2.26.2: If $m \leq n$ and $A \in M_{m,n}, B \in M_{n,m}$ then $x^{n-m} \det(xI_m - AB) = \det(xI_n - BA)$

Theorem (Cayley-Hamilton Theorem): $\chi_A(A) = 0$

Theorem 3.1.1: $AA' = A'A = \det(A)I_n$

Theorem 3.1.2: $A \in \mathrm{GL}_n(F) \Rightarrow A^{-1} = \frac{1}{\det(A)} A'$

Properties:

- A has a zero column $\Rightarrow \det(A) = 0$
- $\det(A^T) = \det(A), \det(\overline{A}) = \overline{\det(A)} = \det(A^*)$
- constant: $\det(A)$
- x^{n-1} term: $-\operatorname{tr}(A)$
- linear term: $(-1)^{n-1} \sum_{j=1}^{n} m_{j,j}$
- x^k term: $(-1)^{n-k} \sum (n-k) \times (n-k)$ principal minors

Elementary Ops and Determinant:

- row $i \Leftrightarrow j$: $\det(A') = -\det(A)$
- $c\times$row i: $\det(A') = c\det(A)$
- $c\times$row $i+$ row j: $\det(A') = \det(A)$

Eigenvalues and Eigenvectors

Spectrum: $\mathrm{Spec}(A) = \{\text{eigenvalues of } A\}$

Eigenspace: $\mathcal{E}_\lambda = \{v \in F^n : Av = \lambda v\}$ (linear subspace of F^n)

Geometric Multiplicity: $\dim(\mathcal{E}_\lambda)$

Algebraic Multiplicity: Multiplicity as a root of $\chi_A(x)$

Algebraically Closed Field: $\forall p \in F[x], p(x) = z_0(x - z_1)\dots(x - z_n), z_i \in F$

Theorem 3.3.2: For $\lambda_1, \dots, \lambda_k$ distinct eigenvalues of A, $v_1 \in \varepsilon_{\lambda_1}, \dots, v_k \in \varepsilon_{\lambda_k}, v_i \neq 0$, the set $\{v_1, \dots, v_k\}$ is linearly independent

Theorem (Spectral Mapping Theorem): F algebraically closed, then $\forall p \in F[x]$, $\mathrm{spec}(p(A)) = p(\mathrm{spec}(A)) := \{p(\lambda) : \lambda \in \mathrm{spec}(A)\}$

Theorem 3.3.4: $\det(A) = \prod_{i=1}^n \lambda_i$ and $\mathrm{tr}(A) = \sum_{i=1}^n \lambda_i$

Concrete vs. Abstract Algebra

Given ordered bases $X = (x_1, \dots, x_n), Y$ of an F-vector space V, W and $v \in V$,

Terminology	Notation
coordinates of v relative to X	$[v]_X = (c_1, \dots, c_n) \in F^n$ s.t. $v = \sum_{i=1}^n c_i v_i$
inverse coordinate map	$L_X : F^n \to V$ via $L_X(c_1, \dots, c_n) = c_1x_1 + \dots + c_nx_n \Leftrightarrow [v]_X = L_X^{-1}(v)$
space of linear maps from V to W	$L(V, W)$
associative F-algebra	$L(V) = L(V, V)$ with composition as multiplication identity $id : V \to V$
T relative to input X and output Y	$M = {}_Y[T]_X$ where $M^{[j]} = L_y^{-1} \circ T \circ L_X(e_j)$ if $X = Y$, ${}_X[T]_X = [T]_X$

$$\begin{tikzcd} V \arrow[r, "T"] & W \\ F^n \arrow[u, "L_X"] \arrow[r, "M"] & F^m \arrow[u, "L_Y"'] \end{tikzcd}$$

Given an n-dim F-vector space V with ordered basis X and m-dim W with Y, the map $\phi : L(V, W) \to M_{m,n}(F)$ via $\phi(T) = {}_Y[T]_X$ is a vector space isomorphism

- ${}_Y[S + T]_X = {}_Y[S]_X + {}_Y[T]_X$
- ${}_Y[cT]_X = c({}_Y[T]_X)$

Example: $X = ((1, 0, 0), (0, 1, 0), (0, 0, 1)), Y = ((1, 3), (-1, 2)), T(a, b, c) = (a+b-c, 3b+2c)$. To compute ${}_Y[T]_X$,

- $L_X(c_1, c_2, c_3) = (c_1, c_2, c_3), L_Y(c_1, c_2) = c_1(1, 3) + c_2(-1, 2) = (c_1 - c_2, 3c_1 + 2c_2)$

- $L_Y^{-1} \circ T \circ L_X(1,0,0) = L_Y^{-1}(1,0) \Rightarrow (c_1 - c_2, 3c_1 + 2c_2) = (1,0) \Rightarrow c_1 = \frac{2}{5}, c_2 = -\frac{3}{5}$
- Repeat for $(0,1,0), (0,0,1)$
- ${}_Y[T]_{X_1} = \begin{bmatrix} \frac{2}{5} & 1 & 0 \\ -\frac{3}{5} & 0 & 1 \end{bmatrix}$

Dual Space

Given $L(V,W)$ is an F-vector space, $S \in L(V,W)$ is a map,

Terminology	Notation
dual space of V	$V^* = L(V,F), V^* \cong V$ for finite dimensional V
dual basis of V^*	$X^* = (f_1, \ldots, f_n)$ s.t. $f_i(x_i) = 1_F, f_i(x_j) = 0_F, \forall i \neq j$
dual map of S	$S^* \in L(W^*, V^*)$ via $S^*(g) = g \circ S$ for $g \in W^*$

Theorem 3.5.2: Given V, W with X, Y, $\forall S \in L(V,W)$, ${}_{X^*}[S^*]_{Y^*} = ({}_Y[S]_X)^T$

Theorem 3.8.1: For three finite dimensional F-vector spaces V, W, U with basis $X, Y, Z, T \in L(V,W), S \in L(W,U) \Rightarrow {}_Z[S \circ T]_X = {}_Z[S]_Y {}_Y[T]_X$

Theorem 3.8.2: $T \in L(V,W)$ has an inverse map $T^{-1} \in L(W,V)$ s.t. $T^{-1} \circ T, T \circ T^{-1}$ are the identity map $\Rightarrow \forall$ ordered bases X of V, Y of W, $({}_Y[T]_X)^{-1} = {}_X[T^{-1}]_Y$

Transition Matrices and Changing Coordinates

Transition Matrix from X to Y: ${}_Y[id]_X$

Column Equivalent: $\exists P \in GL_n(F)$ s.t. $A' = AP$

Row Equivalent: $\exists Q \in GL_n(F)$ s.t. $A' = QA$

Row/column Equivalent: $\exists P, Q \in GL_n(F)$ s.t. $A' = QAP$

Theorem 3.10.1: $\forall$ ordered basis Y of V and every invertible matrix in $\mathrm{GL}_n(F)$ has the form ${}_Y[id]_Z$ for a unique ordered basis Z of V and ${}_X[id]_Y$ for a unique X

Theorem (Projection Theorem): For $T \in L(V,W), \exists$ an ordered basis X' for V and Y' for W s.t. $T(x_i') = y_i', \forall i \in [\mathrm{rank}(T)]$ and $T(x_i') = 0$ for $i > \mathrm{rank}(T)$

Theorem 3.10.4: $L(V) \cong M_n(F)$ as F algebras with isomorphism $T \mapsto [T]_X$ for a fixed ordered basis X of V

Triangulability and Diagonalizability

Similar Matrices:

1. $A' = P^{-1}AP$ for $P \in GL_n(F)$
2. A, A' have the same JCF

Companion Matrix: $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in F[x], C(p) = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{n-1} \end{bmatrix}$

Triangulability:

1. A has n eigenvalues in F counted by algebraic multiplicity $\Rightarrow$ every matrix is triangulable in $M_n(\mathbb{C})$
2. (Strictly) triangulable iff $\exists$ a flag of subspaces (strictly) stabilized

Diagonalizability:

1. F^n has a basis consisting of eigenvectors of A
2. A has n eigenvalues in F counted by geometric multiplicity

Block Triangulability with Two Blocks: F^n has a nontrivial A-invariant subspace

Invariant Subspaces: For $W \subseteq V, T(W) \subseteq W$

- A (complete) flag of subspaces sequence $\{0\} = V_0 \subseteq \cdots \subseteq V_n = V$ s.t. $\dim(V_i) = i$
- A linear map T stabilizes the flag if $T(V_i) \subseteq V_i$ and strictly stabilizes if $T(V_i) \subseteq V_{i-1}$

Direct Sum: $U, W \subseteq V$, if $U \cap W = \{0\}, U \oplus W = \{u + w | u \in U, w \in W\}$

Block Diagonalizability with Two Blocks: $\exists W, Z \subseteq V$ s.t. W, Z are T-invariant subspaces and $V = W \oplus Z$

Projection Map (onto W along Z): $P_{W,Z} : V \to V$ s.t. $P(v) = w$ where $v = w + z$

Theorem 3.12.1: Similar matrices have the same characteristic polynomial, determinant, trace, and minimal polynomial, $\Leftarrow$ false

Theorem 3.24.5: $T \in L(V) \Rightarrow V = \text{img}(T) \oplus \ker(T)$ and T is the projection map of V onto $\text{img}(T)$ along $\ker(T)$

Homework 8: Each eigenspace of A is an A-invariant subspace

Example: $A = \begin{bmatrix} a & b & \dots & b \\ b & a & \dots & b \\ \vdots & \vdots & \ddots & \vdots \\ b & b & \dots & a \end{bmatrix} = (a-b)I_n + bJ_n$ has eigenvectors $v_1 = \mathbf{1}_n$ associated to $a + (n-1)b$ and $v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots$ for $a - b$ which are linearly indept

$P = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \Rightarrow P^{-1}AP = \text{diag}(a + (n-1)b, a - b, \dots, a - b)$

Example: To compute the block triangular matrix similar to A

- compute $\chi_A(x)$
- use Gaussian elimination to compute $\ker(A - \lambda I)$ to obtain a basis for $\mathcal{E}_\lambda$
- extend this to a basis $\mathcal{B}$ of $\mathbb{R}^n$ by computing the orthogonal complements

- let B be the transition matrix from $\mathcal{B}$ to the standard basis
- $B^{-1}AB$ is a block triangular matrix

Bilinear Map

Bilinear Map: $B : V \times V \to F$ is F-bilinear if $\forall w, u, v \in V, c \in F$,

- $B(u, w) + B(v, w) = B(u + v, w)$
- $B(w, u) + B(w, v) = B(w, u + v)$
- $cB(w, u) = B(cw, u) = B(w, cu)$

The space of F-bilinear maps $B : V \times V \to F$ is $BL(V)$ and is a vector subspace of $V \times V \to F$

Congruent Matrices: $\exists P \in GL_n(F)$ s.t. $A' = P^T AP$

Theorem 3.26.4: For a bilinear map $B : V \times V \to F$ and any ordered basis X, Y of V, $[B]_X$ is congruent to $[B]_Y$

Inner Product Spaces and Orthogonality

Inner Product:

- $\mathbb{R}$: $\langle v, w\rangle = w^T v$
- $\mathbb{C}$: $\langle v, w\rangle = w^* v$

Norm: $\|\cdot\| : V \to [0, \infty)$ s.t.

- $\|v\| = 0 \Leftrightarrow v = 0$
- $\forall c \in \mathbb{R}, v \in V, \|cv\| = |c|\|v\|$
- $\forall u, v \in V, \|u + v\| \leq \|u\| + \|v\|$
- if V is an inner product space, the inner product defines a norm by $\|v\| = \sqrt{\langle v, v\rangle}$

Orthogonality: $\langle v, w\rangle = 0$

Orthonormal Basis: $\forall x \neq y \in X, \langle x, x\rangle = 1, \langle x, y\rangle = 0$

Orthogonal Map: $\forall v, v' \in V, \langle T(v), T(v')\rangle = \langle v, v'\rangle$

Orthogonal Group: $O(V) =$ the set of all orthogonal maps in $L(V)$

Orthogonal Matrices:

- $\forall v, w \in \mathbb{R}^n, \langle Av, Aw\rangle = \langle v, w\rangle$
- $AA^T = A^T A = I_n$

Orthogonal Similarity: $\exists$ orthogonal matrix P s.t. $A' = P^{-1}AP = P^TAP$ (A, A' represent the same linear map w.r.t different orthonormal bases)

Theorem (The Orthogonal Matrix Theorem): $A \in M_n(\mathbb{R})$, TFAE:

- A is an orthogonal matrix
- L_A is an orthogonal map
- The rows/columns of A form an orthonormal basis of $\mathbb{R}^n$
- $A^T = A^{-1}$
- A^T is an orthogonal matrix

Matrices with Special Structures

Conjugate Transpose: $A, B \in M_{m,n}(\mathbb{C}), z \in \mathbb{C}$,

- $(A+B)^* = A^* + B^*$
- $(zA)^* = \overline{z}A^*$
- for $D \in M_{n,p}(\mathbb{C}), (AD)^* = D^*A^*$
- $(A^*)^* = A$
- m, n, A invertible $\Leftrightarrow A^*$ invertible
- $\mathrm{tr}(A^*) = \overline{\mathrm{tr}(A)}, \det(A^*) = \overline{\det(A)}$

Quadratic Form: $Q_A(v) = v^*Av = \sum_{i=1}^n \sum_{j=1}^n \overline{v_i} v_j A(i,j)$

Hermitian Matrices:

1. $A^* = A$
2. Q_A is real-valued
3. A is normal and all eigenvalues are real

Skew Hermitian Matrices: $A^* = -A$

Normal Matrices: $AA^* = A^*A$

Unitary Matrices:

- $U^* = U^*U = I_n \Rightarrow U^* = U^{-1}$
- A is normal and all eigenvalues have modulus 1

Positive Definite:

- $\forall v \neq 0 \in \mathbb{C}^n, v^*Av > 0$
- A is normal and all eigenvalues are strictly positive

THEOREM 3.31.1: All eigenvalues of a Hermitian matrix are real, $\Leftarrow$ false

THEOREM 3.31.2: $\forall v \in \mathbb{C}^n, Q_A(v) = 0 \Rightarrow A = 0_{M_n}(\mathbb{C})$

THEOREM (HERMITIAN DECOMPOSITION): Every matrix $A \in M_n(\mathbb{C})$ can be written uniquely as $X + iY$ for X, Y Hermitian in $M_n(\mathbb{C})$

THEOREM 4.2.2: A^*A, AA^* are positive semidefinite. $m = n$ and A is invertible $\Rightarrow A^*A, AA^*$ are positive definite

THEOREM 4.2.3: Positive/negative (semi)definite has positive/nonnegative/negative/nonpositive eigenvalues

THEOREM 4.7.2: TFAE:

- U is unitary
- $\forall v, w \in \mathbb{C}^n, \langle Uw, Uv \rangle = \langle w, v \rangle$
- $\forall v \in \mathbb{C}, \|v\| = \|Uv\|$
- The rows/columns of U are orthonormal
- U^*, U^T is unitary

THEOREM 4.7.3: $U \in U_n(\mathbb{C})$ has all eigenvalues of modulus 1

THEOREM 4.7.4: $\|\det(U)\| = 1$, $\Leftarrow$ false

THEOREM (SPECTRAL THEOREM FOR NORMAL MATRICES): For $A \in M_n(\mathbb{C})$, TFAE:

- A is normal
- A is unitarily diagonalizable
- A admits an orthonormal eigenbasis for $\mathbb{C}^n$
- $\exists$ a polynomial $p \in \mathbb{C}[x]$ s.t. $A^* = p(A)$ (find eigenvalues of A and interpolate them with their conjugates)
- A is normal then A is invertible iff all eigenvalues are nonzero

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PROPERTIES:

- Positive/negative (semi)definite $\Rightarrow$ Hermitian, $\Leftarrow$ false
- The trace of positive/negative definite matrix is positive/negative
- The determinant of positive/negative definite matrix is positive/same sign as $(-1)^n$

(SIMULTANEOUSLY) UNITARY TRIANGULABILITY AND DIAGONALIZABILITY

UNITARY SIMILARITY: $\exists U \in U_n(\mathbb{C})$ s.t. $B = U^*AU = U^{-1}AU$

UNITARILY TRIANGULABLE: : $A \in M_n(\mathbb{C})$ is unitarily triangulable

UNITARILY DIAGONALIZABLE: :

1. A has an orthonormal eigenbasis
2. A is normal

Simultaneously Unitarily Triangulable:

- A, B share an eigenvector $\Rightarrow A, B$ sim. unit. tri.
- A, B sim. unit. tri. $\Rightarrow A, B$ commute ($\Leftarrow$ false) $\Rightarrow A, B$ share an eigenvector

Simultaneously Unitarily Diagonalizable: commute

Summary:

Classes of complex matrices

Name	Definition	Eigenvalues	Quadratic form	Equivalent conditions
Normal matrices	$AA^* = A^*A$	arbitrary	arbitrary	unitarily diagonalizable
Hermitian matrices	$A = A^*$	real	real valued	real-valued quadratic form
Positive definite matrices	positive-valued quadratic form	positive	positive valued	-
Positive semidefinite matrices	nonnegative-valued quadratic form	nonnegative	nonnegative valued	-
Negative definite matrices	negative-valued quadratic form	negative	negative valued	$-A$ is positive definite
Unitary matrices	$A^*A = AA^* = I$	modulus 1	arbitrary	$\langle Uv, Uw\rangle = \langle v, w\rangle$

Every class here is a type of normal matrix and can also be characterized by normality plus the relevant eigenvalue condition. For example, normal plus modulus 1 eigenvalues implies unitary. Additionally every class is closed under unitary similarity.

Theorem (Schur Decomposition): Every matrix $A \in M_n(\mathbb{C})$ is unitarily triangulable

Theorem 4.7.7: Every set of pairwise orthogonal, unit normed vectors can be extended to an orthonormal basis

Theorem 4.12.1: $A, B \in M_n(\mathbb{C}) \Rightarrow \forall p(x) \in \mathbb{C}[x], B$ commutes with $p(A)$

Theorem (Lagrange Interpolation Formula): For any field $F, n \in \mathbb{N}$, given distinct $a_0, \ldots, a_n \in F$ and any (not necessarily distinct) $b_0, \ldots, b_n \in F$, there exists a unique polynomial $p \in F[x]$ of degree at most n s.t. $p(a_i) = b_i, \forall 0 \leq i \leq n$

Theorem 4.12.3: For every positive definite (general) matrix $A \in M_n(\mathbb{C}), k \geq 2$ even, $\exists$ unique positive (semi)definite B s.t. $B^k = A$; for every Hermitian $A \in M_n(\mathbb{C}), k \geq 1$ odd, $\exists$ a unique Hermitian matrix B s.t. $B^k = A$.

Example: Compute Schur Decomposition of A:

- find an eigenvector of A and extend it to an orthonormal basis of $\mathbb{C}^n$
- let B_1 be the transition matrix
- compute $A = B_1^{-1}AB_1$
- repeat the process until A has one nonzero column
- let $U = \prod_i B_i \Rightarrow U^*AU$ is upper-triangular

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Matrix Decomposition

Polar Decomposition: $\forall A \in GL_n(\mathbb{C}), \exists$ unique positive definite matrix R and $U \in U_n(\mathbb{C})$ s.t. $A = RU$

UR Decomposition / Dual Polar Decomposition: $\exists$ unique R positive semidefinite and (not necessarily distinct) $U \in U_n(\mathbb{C})$ s.t. $A = UR$

If A is normal, then $UR = RU$

Singular Value Decomposition: $\exists P, Q \in U_n(\mathbb{C})$ and D diagonal with nonnegative entries s.t. $A = PDQ^*$

D is uniquely determined up to the order of diagonal entries and the entreis are SINGULAR VALUES of A

EXAMPLE: Compute UR decomposition:

- if A is symmetric, it suffices to compute the polar decomp
  - $R = \sqrt{AA^*}, U = R^{-1}A$
- else $R = \sqrt{AA^*}$ and R may be related to A by permuting rows

EXAMPLE: Compute SVD decomposition:

- unitarily diagonalize R and get $D = Q^*RQ$; D has diagonal entries as eigenvalues of R and Q has columns as normalized eigenvectors of R
- $P = UQ \Rightarrow A = PDQ^*$

JORDAN CANONICAL FORM

JORDAN BLOCK: $J(c;m) = \begin{bmatrix} c & 1 & 0 & \dots & 0 \\ 0 & c & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & c \end{bmatrix} \in M_n(\mathbb{C})$

JORDAN CANONICAL FORM: $J = \text{blk-diag}(J(c_1;m_1), \dots, J(c_s;m_s))$

NILPOTENT MATRICES: $\exists k \in \mathbb{N}$ s.t. $A^k = \mathbf{0}_{M_n(F)} \Rightarrow 0$ is an eigenvalue

INDEX OF NILPOTENCE: smallest k

MINIMAL POLYNOMIALS: $\exists$ unique monic polynomial m_A of minimal degree s.t. $m_A(A) = 0$

THEOREM (JORDAN CANONICAL FORM THEOREM): Given any linear map T on a finite dimensional $\mathbb{C}$-vector space, $\exists$ an ordered basis X of V s.t. $J = [T]_X$ is a Jordan Canonical Form

JCF is unique up to reordering of Jordan blocks

THEOREM (FITTING'S LEMMA): Given an n-dimensional F-vector space V and $T \in L(V), \exists T$-invariant subspaces Z, W of V s.t. $V = Z \oplus W$ with $T|_Z$ nilpotent and $T|_W$ an isomorphism

THEOREM 4.26.1: $U \in L(V)$ s.t. $\ker(U) = \ker(U^2) \Rightarrow V = \ker(U) \oplus \text{img}(U)$

THEOREM 4.28.3: A is a block diagonal matrix in $M_n(F)$ with blocks $A_1, \dots, A_s$ then $m_A(x) = \text{lcm}(m_{A_1}(x), \dots, m_{A_s}(x))$

THEOREM 4.28.4: $p(A) = 0$ iff $m_A(x)$ divides $p(x)$

THEOREM 4.30.1: The minimal polynomial of $J(c;k)$ is $(x-c)^k$

THEOREM (CAYLEY-HAMILTON THEOREM): $\chi_A(A) = \mathbf{0}_{M_n(\mathbb{C})}$; for $A \in M_n(F), m_A(x)$ divides $\chi_A(x)$

THEOREM 4.30.2: If $A \in M_n(\mathbb{C})$ is diagonalizable then $m_A(x) = (x-\lambda_1)\dots(x-\lambda_s)$ for $\text{Spec}(A) = \{\lambda_1, \dots, \lambda_s\}$

EXAMPLE: Compute JCF of $A = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$:

- if A is Hermitian $\Rightarrow$ unitarily diagonalizable $\Rightarrow$ JCF is diagonal with eigenvalues as entries

- compute $\chi_A(x) = (x-1)^2(x+1)^2$
- find the Jordan block for -1: $\dim(\ker(A+I)) = 1, \dim(\ker((A+I)^2)) = 2, \dim(\ker((A+I)^3)) = 2 \Rightarrow \mu'_{-1} = (1,1) \Rightarrow J(-1;(2))$
- find the Jordan block for 1: $J(1;(2))$
- $J_A = \text{blk-diag}(J(-1,(2)), J(1,(2)))$

Spectral Graph Theory

Adjacency Matrix: $A(G)(i,j) = \begin{cases} 1 & \text{if } ij \in E \\ 0 & \text{otherwise} \end{cases}$

Eigenvalues of a Graph: Eigenvalues fo $A(G)$ are real; A admits an orthonormal eigenbasis of $\mathbb{R}^n$ Eigenvalues of Bipartite Graph: The spectrum of a bipartite graph is symmetric about zero

Incidence Matrix: $M(G) \in M_{n,m}(\mathbb{R})$ with rows indexed by vtxs and columns indexed by edges s.t. $M(v,e) = \begin{cases} 1 & \text{if } e \text{ is incident to } v \\ 0 & \text{otherwise} \end{cases}$

Boundary Matrix: $\partial(G) \in M_{n,m}(\mathbb{R})$ s.t. $\forall ij \in E, i < j, M(i,e) = 1, M(j,e) = -1, M(k,e) = 0, \forall k \neq i,j$

Laplacian: $\mathcal{L}(G) = \partial\partial^* \in M_n(\mathbb{C})$ positive semidefinite $\Rightarrow$ real, nonnegative eigenvalues

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Theorem 5.3.1: $A^k(i,j)$ counts # walks of length k from i to j

Theorem 5.3.2: The diameter of a connected graph G is less than its number of distinct eigenvalues

Theorem 5.3.3: For any graph G with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$, we have $\lambda_1 \geq \Delta(G)$ with equality iff some component of G is $\Delta(G)$-regular

Theorem 5.3.4: $|\lambda| \leq \Delta(G)$

Theorem 5.3.5: For a connected graph G, $-\Delta(G)$ is an eigenvalue iff G is $\Delta(G)$-regular and bipartite

Theorem 5.5.1: Let D be the degree matrix of G (diagonal entries are $\deg(i)$) $\Rightarrow \mathcal{L} = D - A$

Theorem (Kirchhoff's Matrix Tree Theorem): For a graph G with $0 \leq \lambda_1 \leq \cdots \leq \lambda_n$ as eigenvalues of $\mathcal{L}(G)$, the number of labeled spanning tree of G is $\frac{\lambda_2 \ldots \lambda_n}{n}$

Theorem 5.5.2: For any graph $G = (V,E)$ with $|E| = |V| - 1, \det(\partial(\hat{G})) \pm 1$ iff G is a tree

Theorem (Enumerating Spanning Trees in a Regular Graph): G is d-regular and the eigenvalues of the adjacency matrix are $d = \lambda_1 \geq \cdots \geq \lambda_n$, then the eigenvalues of the Laplacian are $d - \lambda_i, i \in [n]$

Example: $A(K_n) = J_n - I_n$: the eigenvectors are $\mathbf{1}_n$ associated to $n-1$ and $v_j = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ -1 \\ \vdots \\ 0 \end{bmatrix}, j = 2, \ldots, n$ associated to -1 with geometric multiplicity $n-1$

Review Problems

- $\left[\begin{smallmatrix} 1 & i \\ 0 & 1 \end{smallmatrix}\right]$: not normal, $p(A) = (1+i)A - iA = A^{-1}$, eigenvalue 1
- $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right]$: not unitarily diagonalizable
- $\left[\begin{smallmatrix} i & 1 \\ 0 & 1 \end{smallmatrix}\right]$: A^{-1} is similar to A^* but A is not similar to a unitary matrix